

Inhomogeneity, magnetic auto-Bäcklund transformations and magnetic solitons for a generalized variable-coefficient Kraenkel-Manna-Merle system in a deformed ferrite

Xin-Yi Gao ^{a, b}  , Jian-Guo Liu ^c  , Gang-Wei Wang ^d   [Show more](#) ▾ Cite  Add to Mendeley  Share | 10.1016/j.aml.2025.109615 ↗[Access through your organi...](#)[Purchase PDF](#)[Article preview](#)[Recommended articles](#)[Cited by \(219\)](#)[Metrics](#)

Abstract

Almost all the modern electronic devices have the ferromagnetic parts. For a generalized variable-coefficient Kraenkel-Manna-Merle system in a deformed ferrite, which is a generalized variable-coefficient ultrashort wave model, this paper, around a noncharacteristic movable singular manifold, symbolically computes out two magnetic auto-Bäcklund transformations, along with two families of the magnetic soliton solutions. Those results are in their reliance on the coefficients in that system. Electrodynamical implications of our magnetic auto-Bäcklund transformations and magnetic solitons come from the follow factors: the magnetic field, magnitude of the density of saturation magnetization, magnetization density, vacuum velocity of light, gyromagnetic ratio, shape of the wave during the propagation with a very short wavelength assumed, slow time variable describing the propagation over a long time or a long distance linked to the wavelength, inhomogeneity in the ferrite measuring the bond dependence of lattice defects and the corresponding exchange effects (also known as the system deformation) as well as exchange integral between the nearest-neighbor spin-spin interaction for the ferrite. With symbolic computation, the impact of inhomogeneity on the magnetic auto-Bäcklund transformations and on the transmission of magnetic solitons is presented. On the application side, the inhomogeneity could offer the possibility to realize the ultrafast magnetization switching in certain magnetic devices. Future experiments and observations might detect some features predicted in this paper, and relevant physical insights might be expected.

Keywords

Ferrite; Inhomogeneity; Generalized variable-coefficient Kraenkel-Manna-Merle system; Magnetic auto-Bäcklund transformation; Magnetic soliton; Singular manifold; Symbolic computation

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Introduction

Nowadays, a ferromagnetic material, such as a ferrite, has been known as a material with a spontaneous magnetization and employed in the electronic and energy industries since almost all the modern electronic devices have the ferromagnetic parts [1], [2], [3], [4], [5]. Of interdisciplinary interest, nonlinear studies have also been active [6], [7], [8], [9], [10], [11], [12], [13], [14], [15], [16], [17], [18], [19], [20], [21], [22], [23], [24], [25], [26], [27].

Correspondingly, people have studied many Kraenkel-Manna-Merle (KMM)-type models, such as certain nonlinear ultra-short wave pulses in a saturated ferromagnetic material including the external field but excluding the conductivity through a KMM system in the fractional differential sense [28], rogue waves for a KMM system on a periodic background [29], Darboux transformation for a KMM integrable system characterizing the nonlinear dynamics of ultra-short wave pulses within a saturated ferromagnetic material [30], solitons in a saturated ferromagnetic material modeled via a fractional KMM system [31], invariant solutions, Lie symmetry analysis, bifurcations and nonlinear dynamics for a KMM system with/without the damping effect [32], complex dynamics and solitons for a KMM model describing a saturated ferromagnetic material [33], influence of the internal magnetic field on the magnetization in a ferromagnetic material via a KMM system [34], dynamical study of certain analytic solutions, bifurcation analysis and chaotic nature for a complex KMM system [35], solitons and other solutions for a space-time fractional KMM model [36], localized wave solutions for a KMM system by virtue of the Darboux transformation, Lax pair and neural network [37], dynamical study and soliton interactions for a KMM system in a ferrite [38], N -fold Darboux transformation, generalized Darboux transformation and solitons for a complex KMM system in a ferrite [5], similarity solutions for a (2+1)-dimensional KMM system with the Gilbert damping and inhomogeneous exchange parameters in a saturated ferromagnetic film [39], closed-form solitons for a fractional stochastic KMM system [40], N -fold Darboux transformation and soliton interactions for a KMM system in a saturated ferromagnetic material [41], generalized Darboux transformation and solitons for a KMM system in a ferromagnetic saturator [42], dynamics of the damped and undamped waves for a fractional KMM system in a ferromagnetic material [43], noise effect on the soliton phenomena for a fractional stochastic KMM system in a ferromagnetic material [44], traveling wave solutions with the qualitative behaviors for a fractional stochastic KMM equation in a ferromagnetic material [45], solitons for a fractional stochastic KMM system in a ferromagnetic material [46], solitons for a fractional KMM model [47], analytic solutions for a fractional KMM system in a ferromagnetic material [48], solitons for a KMM system with the beta time derivative [49], wave-propagation dynamics for an integrable KMM system under the zero damping effect in a ferrite material [50] as well as lump, rogue, M -shaped, breather-lump and kink-cross rational waves for a KMM system in a saturated ferromagnetic material [51].

Recently, Ref. [52], with references therein, has considered a deformed ferrite slab lying on the (X, T) plane with the negligible transverse dimension. The ferrite has been magnetized to saturation $\vec{M}_0$ by an external field $\vec{H}_0^\infty$ along the Y axis. Electromagnetic waves have possibly propagated along the X axis. For the magnetic field $\vec{H}$ and magnetization density $\vec{M}$, Ref. [52] has made use of the Maxwell and Landau-Lifschitz

equations along with the multi-scale expansion and short-wave approximation, so as to investigate the magnetic solitons and ultrashort waves in the ferrite, and to accordingly propose a generalized variable-coefficient KMM system¹ as

$$\mathcal{C}_{xt} = -\mathcal{B}\mathcal{B}_x ,$$

$$\mathcal{B}_{xt} = \mathcal{B}\mathcal{C}_x + \mathcal{H} [\mathcal{F}'(x) \mathcal{B}_x + \mathcal{F}(x) \mathcal{B}_{xx}] ,$$

where

$$x = -\frac{m}{2\epsilon}(X - VT) , \quad t = m\epsilon T , \quad \mathcal{H} = -\frac{1}{4}J_2m^2 ,$$

the “ ’ ” sign represents the differentiation in respect of x , T is the scaled temporal coordinate with respect to the vacuum velocity of light, ϵ is a small parameter related to the magnitude of the wavelength, V describes the velocity of the short wave, m is the magnitude of $\vec{M}_0$ (the density of saturation magnetization), ϵT is a slow time variable describing the propagation over a long time or a long distance linked to the wavelength, $\frac{1}{\epsilon}(X - VT)$ is linked to the shape of the wave during the propagation with a very short wavelength assumed, $\mathcal{B}(x, t)$ is related to the magnitude of the density of saturation magnetization, the first-order perturbation of the magnetization density, the vacuum velocity of light and the gyromagnetic ratio, $\mathcal{C}(x, t)$ is related to the magnitude of the density of saturation magnetization, the zeroth-order perturbation of the magnetic field, the vacuum velocity of light and the gyromagnetic ratio, J_2 is the second-order perturbation of the exchange integral between the nearest-neighbor spin-spin interaction for the ferrite, as well as $\mathcal{F}(x)$ denotes the inhomogeneity in the ferrite with the inhomogeneity measuring the bond dependence of lattice defects and the corresponding exchange effects, also known as the system deformation. More details can be seen in Ref. [52] and references therein.

System (1) has been said to be non-integrable and used to describe the propagation of magnetic solitons in that deformed ferrite [52]. Numerical simulations for System (1) have been performed to examine certain impacts of the inhomogeneities on the transmission of magnetic solitons [52]. For the homogeneous case, i.e., $\mathcal{F}(x) = 0$, System (1) has been known to reduce to the so-called ideal KMM system [52] (and references therein).

Nevertheless, to our knowledge, for System (1), there has been no Bäcklund-transformation contribution reported until now. Employing symbolic computation [41], [53], we aim to build some magnetic auto-Bäcklund transformations and magnetic solitons.

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Magnetic auto-Bäcklund transformations for System (1)

For System (1), we come up with the following truncated Painlevé expansions in the generalized Laurent series [54], [55]:

$$\mathcal{C}(x, t) = \lambda^{-\Omega}(x, t) \sum_{\omega=0}^{\Omega} \mathcal{C}_{\omega}(x, t) \lambda^{\omega}(x, t) \quad ,$$

$$\mathcal{B}(x, t) = \lambda^{-\Xi}(x, t) \sum_{\xi=0}^{\Xi} \mathcal{B}_{\xi}(x, t) \lambda^{\xi}(x, t) \quad ,$$

around a noncharacteristic movable singular manifold identified with an analytic function $\lambda(x, t) = 0$, where $\mathcal{C}_{\omega}(x, t)$'s and $\mathcal{B}_{\xi}(x, t)$'s are also the analytic functions, $\mathcal{C}_0(x, t) \neq 0$, $\mathcal{B}_0(x, t) \neq 0$, $\lambda_x(x, t) \neq 0$, and the positive integers

$$\Omega = \Xi = 1$$

due to the balance among the powers of λ at the lowest orders.

Next, with ...

Magnetic solitons for System (1)

Keeping the explicitly-solvable goal in mind, we substitute the assumptions

$$\lambda(x, t) = e^{\eta_1(x)t + \eta_2(x)} + 1 \quad ,$$

$$\mathcal{C}_1(x, t) = \eta_3(x)t + \eta_4(x) \quad ,$$

$$\mathcal{B}_1(x, t) = \eta_5(x)t + \eta_6(x) \quad ,$$

into Magnetic Auto-Bäcklund Transformations (2)–(11), to give a try, by virtue of symbolic computation, in which $\eta_1(x), \dots, \eta_6(x)$ are the real differentiable functions of x , and $\eta_1'(x)t + \eta_2'(x) \neq 0$. That way,

$$(\lambda^0 t^2, \lambda^0 t^2) : \eta_5(x) = 0 \quad (\text{as chosen}) \quad ,$$

$$(\lambda^0 t^0, \lambda^0 t^1) : \eta_3(x) = \eta_3 = \text{constant} \quad ,$$

$$\eta_6(x) = \eta_6 = \text{constant} \quad ,$$

$$\left(- , \lambda^0 t^0 \right) : \eta_4(x) = \eta_4 = \text{constant} \quad \text{with} \quad \eta_6 \neq 0 \quad ,$$

$$\left(\lambda^{-2} t^1 , - \right) : \eta_1(x) = \eta_1 = \text{non-zero constant} \quad \text{with} \quad \textcolor{red}{msubsup}$$

...

Conclusions

For System (1), a generalized variable-coefficient KMM system in a deformed ferrite, this paper, around a noncharacteristic movable singular manifold, has symbolically computed out Magnetic Auto-Bäcklund Transformations (2)–(11) along with Magnetic Soliton Solutions (22).

Those results have been in their reliance on $\mathcal{F}(x)$ and $\mathcal{H}$, the coefficients in System (1). Electrodynamical implications of Magnetic Auto-Bäcklund Transformations (2)–(11) and Magnetic Solitonic Solutions (22) have come from the ...

Acknowledgments

The authors express sincere thanks to the Editors and Reviewers for the valuable comments. This work has been supported by the Scientific Research Foundation of North China University of Technology under Grant Nos. 11005136025XN076-092 and 11005136024XN147-87, Natural Science Foundation of Hebei Province of China (No. A2023207002), “333 Talent Project” of Hebei Province (No. C20221021), Key Program of Hebei University of Economics and Business (2023ZD10, 2024ZD12) and Youth Team Support Program ...

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